Departement Elektriese, Elektroniese en Rekenaar-Ingenieurswese

Department of Electrical, Electronic and Computer Engineering

Semestertoets 1

Semester Test 1

Module EBN111

Elektrisiteit en Elektronika

3 Maart 2016

Module EBN111

Electricity and Electronics

3 March 2016

UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting: *Exam information:* Memo			
Maksimum punte: *Maximum marks:*	46	Volpunte: *Full marks:*	45
Duur van vraestel: *Duration of paper:*	90 minute *90 minutes*	Oopboek / toeboek: *Open / closed book:*	Toe *Closed*
Totale aantal bladsye (hierdie blad ingesluit): *Total numer of pages (including this page):*		16 + Answer sheet	

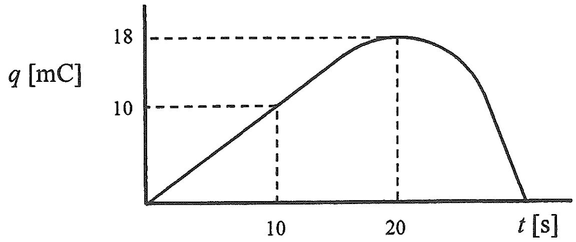
BELANGRIK- *IMPORTANT*

1. Die eksamenregulasies van die Universiteit van Pretoria geld.
   The examination regulations of the University of Pretoria apply.
2. Alle vrae moet op die ANTWOORDBLAD beantwoord word.
   All questions must be answered on the ANSWER SHEET.
3. Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
   Students may do their own rough calculations on the question paper – the question paper will not be taken in.
4. Antwoorde moet eenhede insluit!
   Answers must include units!
5. **Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.**
   Final answers must be written as real numbers, and not as fractions.

Dosent(e): *Lecturer(s):*	1. G Hancke
	2. D. Bhatt
	3. D. le Roux

VRAAG/QUESTION 1 [12]

Vraag 1.1 [2]	Question 1.1 [2]
Die lading wat in 'n geleier vloei word onder getoon. Bepaal die stroom by: a) t =10 s en b) t = 20 s. (Die kurwe bereik sy maksimumwaarde by t = 20 s.)	The charge flowing in a conductor is shown below. Determine the current at: a) t = 10 s en b) t = 20 s. (The curve reaches its maximum value at t = 20 s.)



$$i(10s) = \frac{dq}{dt} = \frac{10mC}{10s} = 1mA$$

$$i(20s) = \frac{dq}{dt} = 0 \ (\text{Slope} = 0)$$

Vraag 1.2 [2]	Question 1.2 [2]
Die spanning oor 'n 5 Ω weerstand neem liniêr toe van 0 mV tot 15 mV tussen t = 0 s en t = 20 s. Bereken a) Die waarde van die lading wat deur die weerstand vloei tussen t = 5 s en t = 15 s. b) Die drywing in die weerstand geabsorbeer by t = 12 s.	The voltage across a 5 Ω resistor increases linearly from 0 mV to 15 mV between t = 0 s and t = 20 s. Calculate a) The value of the charge flowing through the resistor between t = 5 s and t = 15 s. b) The power absorbed by the resistor at t = 12 s.

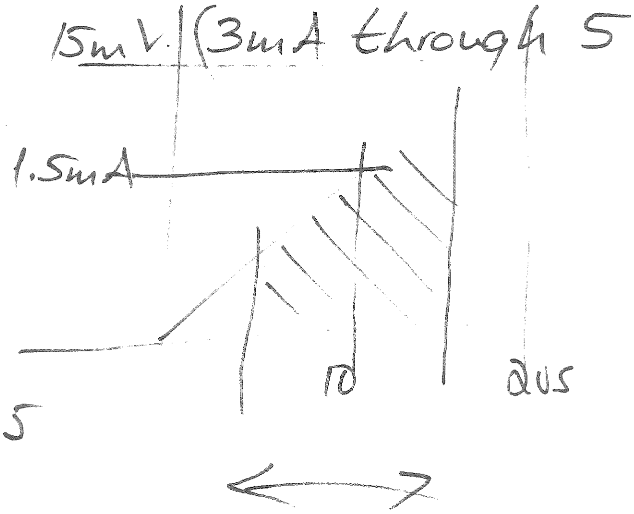
(b) At $t = 12s$

$i = \frac{12}{20} \times 3mA$

$= 1{,}8mA$

$P(12s) = (1.8 \times 10^{-3})^2 \times 5$

$= 16{,}2 \mu W$



(a) area $= 1.5mA \times 10$

$= 15mC$

Vraag 1.3 [3]	Question 1.3 [3]
'n Elektriese verwarmer is gespesifiseer as 240 V; 3 kW. a) Hoeveel stroom trek die verwarmer? b) Hoeveel lading vloei deur dit in 1 minuut? c) Elektrisiteit kos R3/kWh. Bereken die totale koste oor een maand (30 dae) indien die verwarmer elke oggend vir 80 minute aangeskakel word?	An electric heater is specified as 240 V; 3 kW. a) What is the current flowing through the heater? b) How much charge flows through the heater in 1 minute? c) Electricity cost R3/kWh. Calculate the total cost if the heater is switched on for 80 minutes every morning for 1 month (30 days).

(a) $i = \frac{3000}{240} = 12.5\ A$

(b) $q(1\ \text{minute}) = 12.5 \times 60$

$= 750\ C$

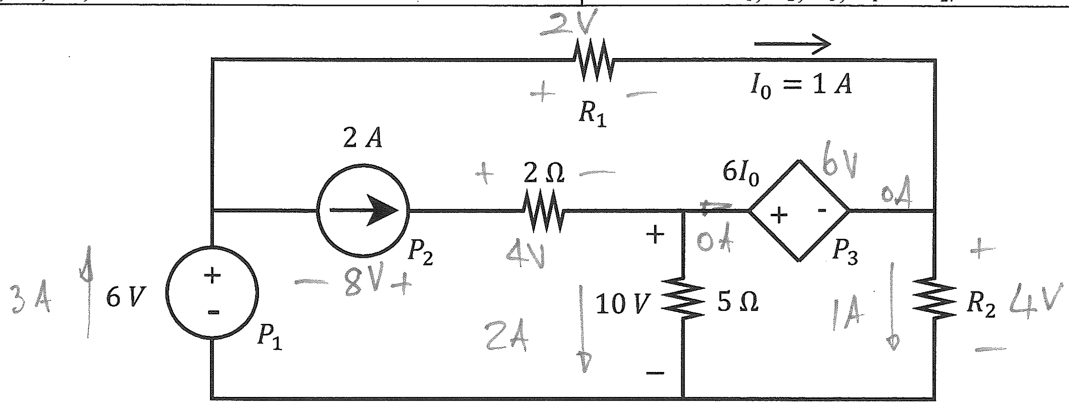
(c) Total energy $= \frac{30 \times 80}{60} \times 3$

$= 120\ kWh$

Total Cost $= 120 \times 3$

$= R360$

Vraag 1.4 [5]	Question 1.4 [5]
Bepaal P_1, P_2, P_3, R_1 en R_2	Determine P_1, P_2, P_3, R_1 en R_2.



$P_1 = -18\,W \; (-3 \times 6)$

$P_2 = -16\,W \; (-2 \times 8)$

$P_3 = 0\,W$ (zero current)

$\sum P_{source} = -34\,W$

$R_1 = 2\,\Omega$

$R_2 = 4\,\Omega$

$P_{R1} = 2\,W$

$P_{R2} = 4\,W$

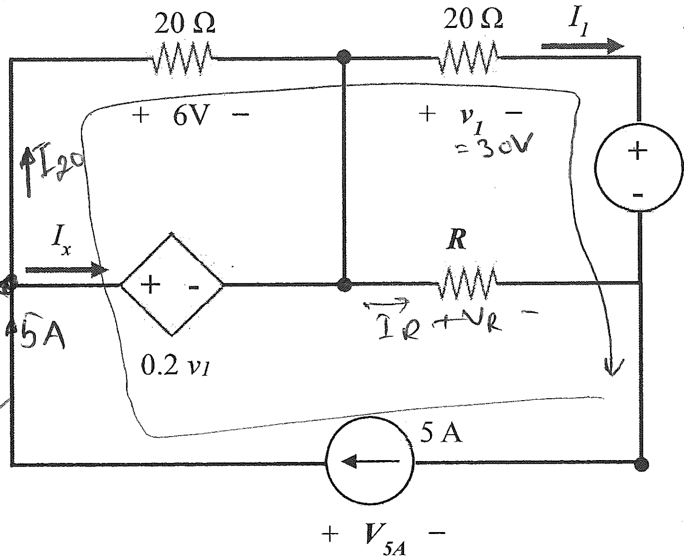
$P_{2\Omega} = 8\,W$

$P_{5\Omega} = 20\,W$

$\sum P_R = +34\,W$

VRAAG/QUESTION 2 [18]

Vraag 2.1 [4]	Question 2.1 [4]
Vir onderstaande kringbaan, bereken die volgende: a) die stroom I_1, b) die stroom I_x, c) die spanning oor die 5 A stroombron, V_{5A}, en d) die waarde van die resistor R wat sal veroorsaak dat 'n stroom van 3.5 A daardeur vloei.	Consider the circuit diagram shown below and calculate the following: a) the current I_1, b) the current I_x, c) the voltage across 5 A current source, V_{5A}, and d) the value of resistor R, that will allow a current of 3.5 A to flow through it.



a) $I_1 = \frac{V_1}{20}$

$0.2V_1 = 6$

$\therefore V_1 = \frac{6}{0.2} = 30V$

$\Rightarrow I_1 = \frac{30}{20} = 1.5A$

b) $I_{in} = I_{out}$

$5 = I_x + I_{20}$

$I_x = 5 - \frac{6}{20}$

$= \frac{100-6}{20}$

$I_x = 4.7A$

c) KVL: $-V_{5A} + 6 + V_1 + 40 = 0$

$V_{5A} = 6 + 30 + 40$

$V_{5A} = 76V$

d) $R = \frac{V_R}{I_R} = \frac{V_R}{3.5}$

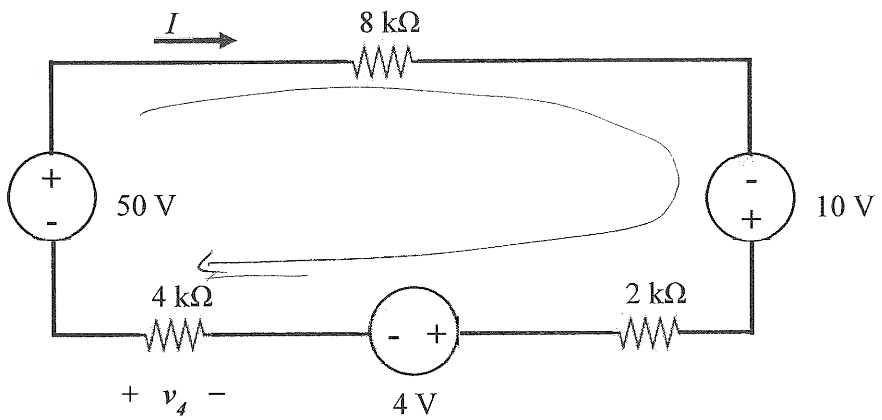
$-V_5 + 0.2V_1 + V_R = 0$

$V_R = 76 - 0.2 \times 30$

$= 70V$

$\therefore R = \frac{70}{3.5} = 20\Omega$

Vraag 2.2 [4]	**Question 2.2 [4]**
Bepaal die: a) spanning oor 4 kΩ resistor, v_4. b) stroom I.	Determine the: a) voltage across 4 kΩ resistor, v_4. b) current I.



(a) using voltage divider

$$v_4 = -(50+10-4)\left(\frac{4k}{(4+8+2)k}\right)$$

$$v_4 = \underline{-16V}$$

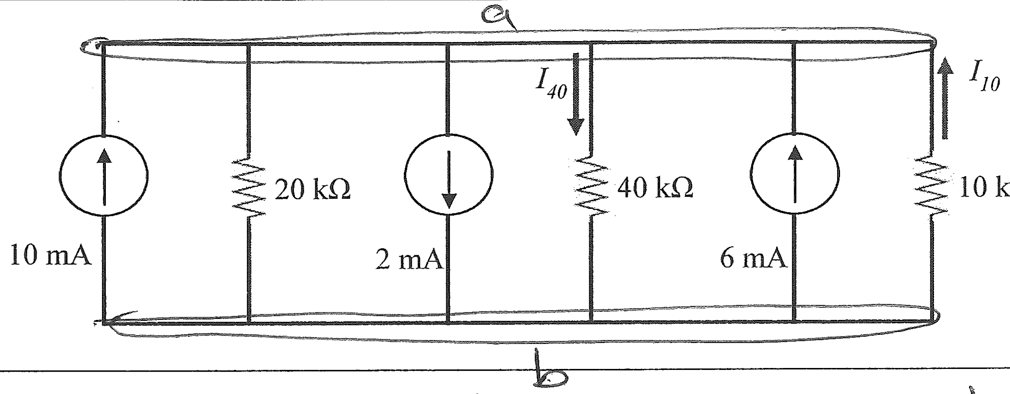
(b) $I = \frac{V}{R}$

$$= \frac{50+10-4}{(8+2+4)k}$$

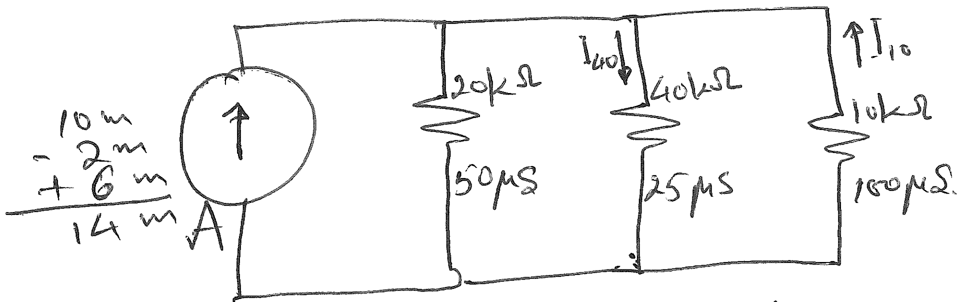
$$I = \frac{56}{14k}$$

$$= \underline{4mA}$$

Vraag 2.3 [4]	Question 2.3 [4]
Bepaal die stroom deur a) die 40 kΩ resistor, I_{40} and b) die 10 kΩ resistor, I_{10}.	Determine the current flowing through a) the 40 kΩ resistor, I_{40} and b) the 10 kΩ resistor, I_{10}.



Note all elements are parallel.



Using current divider

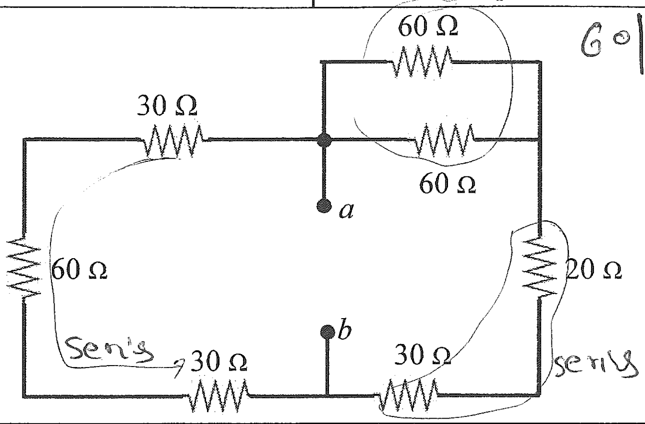
a) $I_{40} = \frac{G_{40}}{G_{eq}} \times I = \frac{1/40k}{1/40k + \frac{1}{20k} + \frac{1}{10k}} \times I = \frac{25\mu}{(50+25+100)\mu} \times 14m$

$I_{40} = \frac{25}{175} \times 14m = \underline{2mA}$

b) $I_{10} = -I \times \frac{G_{10}}{G_{eq}} = -14m \times \frac{100\mu}{175\mu}$

$I_{10} = \underline{-8mA}$

Vraag 2.4 [2] Bepaal die ekwivalente weerstand R_{ab}.	Question 2.4 [2] Calculate the equivalent resistance R_{ab}.



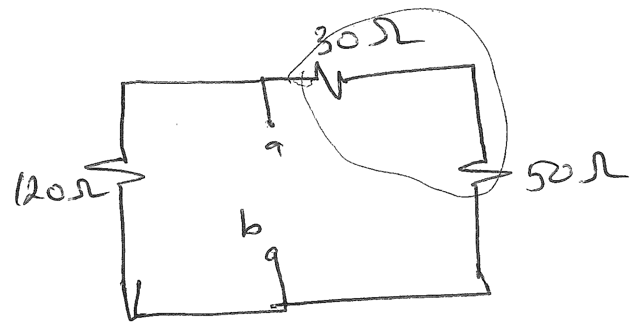
$60 \| 60 = \frac{60 \times 60}{60+60}$

$= 30\,\Omega$

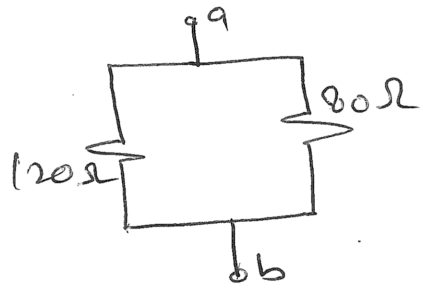
$30+60+30$

$= 120\,\Omega$

$30+20 = 50\,\Omega$



$30+50 = 80\,\Omega$

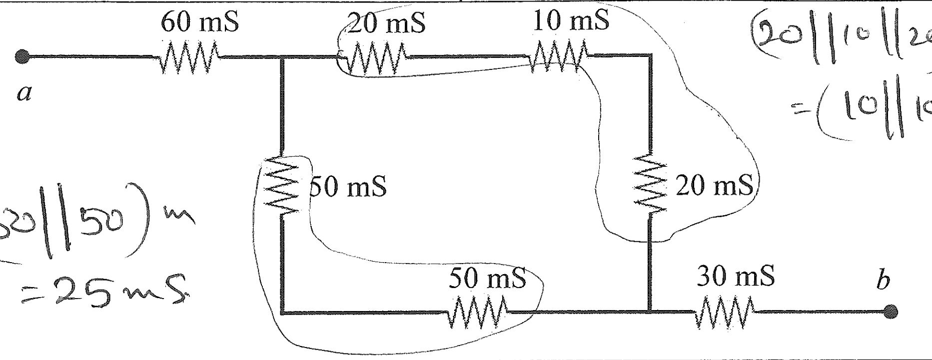


$R_{ab} = 80 \| 120$

$= \frac{80 \times 120}{80+120}$

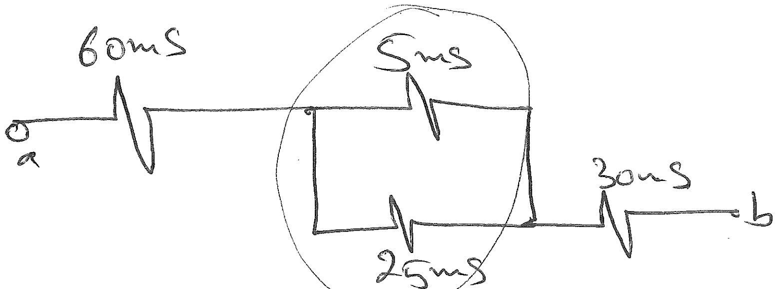
$= \underline{48\,\Omega}$ ✓

Vraag 2.5 [2]	Question 2.5 [2]
Bepaal die ekwivalente geleiding, G_{ab}, tussen die nodes a-b vir onderstaande kringbaan.	Calculate the equivalent conductance, G_{ab}, across the nodes a-b of the circuit below.



$(20 \| 10 \| 20)$ mS
$= (10 \| 10) = 5$ mS

$(50 \| 50)$ m
$= 25$ mS



$(5+25)$ mS
$= 30$ mS.

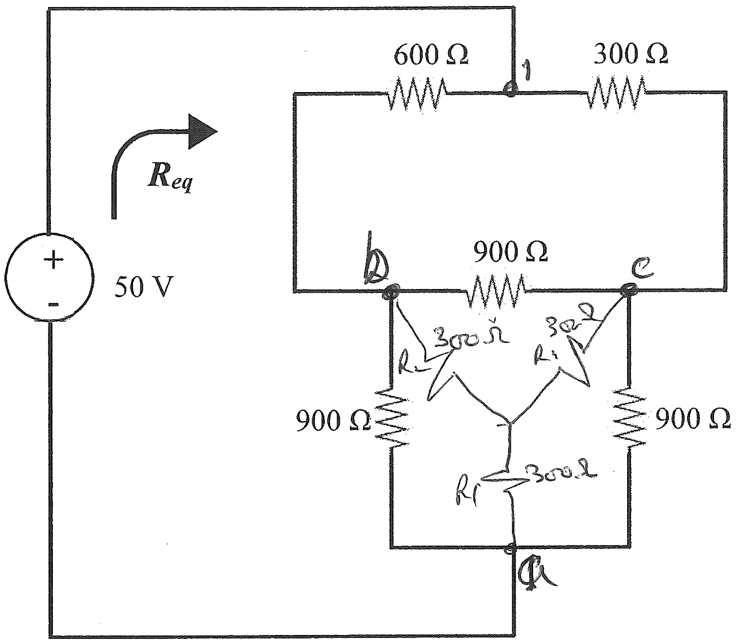
$\Rightarrow G_{ab} = (60m \| 30m \| 30m)$ S

$= (60m \| 15m)$ S

$= \left(\frac{60 \times 15}{60+15}\right)$ mS

$= 12$ mS

Vraag 2.6 [2]	**Question 2.6 [2]**
Bepaal die ekwivalente weerstand R_{eq}.	Calculate the equivalent resistance, R_{eq}.



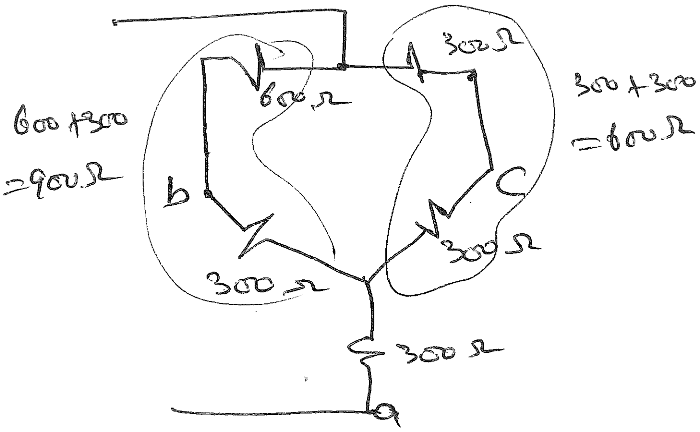
Balanced load

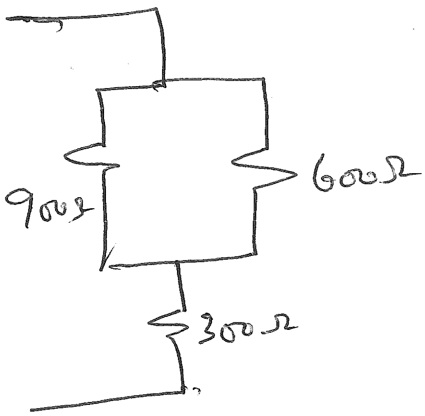
$R_Y = \frac{R_\Delta}{3}$

$= \frac{900}{3}$

$R_{Y_1} = 300\Omega$

$R_1 = R_2 = R_3$





$\therefore R_{eq} = 300 + (900 // 600)$

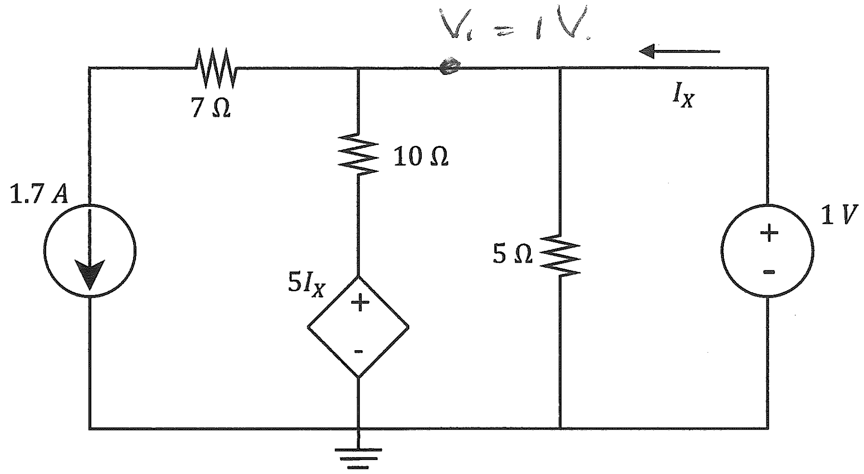
$= 300 + \frac{900 \times 600}{900 + 600}$

$= 300 + 360$

$= 660\Omega$

VRAAG/QUESTION 3 [16]

Vraag 3.1 [2]	Question 3.1 [2]
Gebruik of node of lusanalise en bepaal I_X.	Use either nodal or mesh analysis to determine I_X.



① $I_x = \frac{1}{5} + \frac{1-5I_x}{10} + 1.7.$

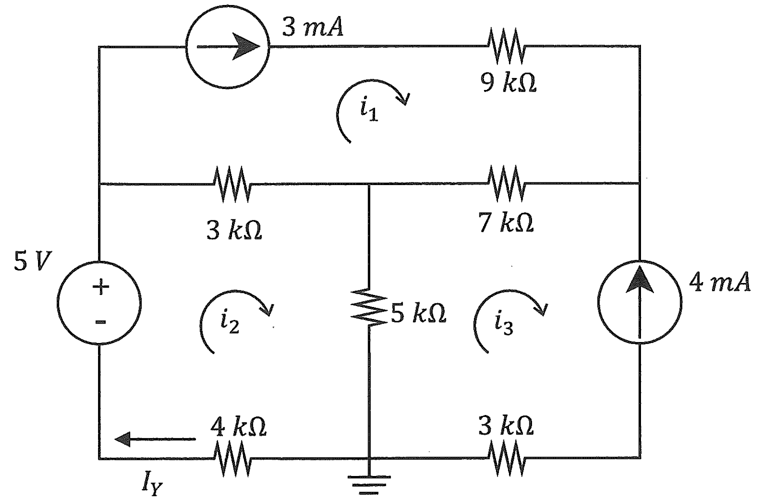
$10I_x = 2 + 1 - 5I_x + 17.$

$15I_x = 20$

$3I_x = 4$

$I_x = \frac{4}{3} = 1.333\ A$

Vraag 3.2 [2]	Question 3.2 [2]
Gebruik lusanalise en bepaal I_Y. (Wenk: i_1= 3 mA, i_2= I_Y, en i_3 = - 4 mA.)	Use mesh analysis to determine I_Y. (Hint: i_1= 3 mA, i_2= I_Y, and i_3 = - 4 mA.)



① $-5 + i_2(3k + 5k + 4k) - 3k\, i_1 - 5k\, i_3 = 0.$

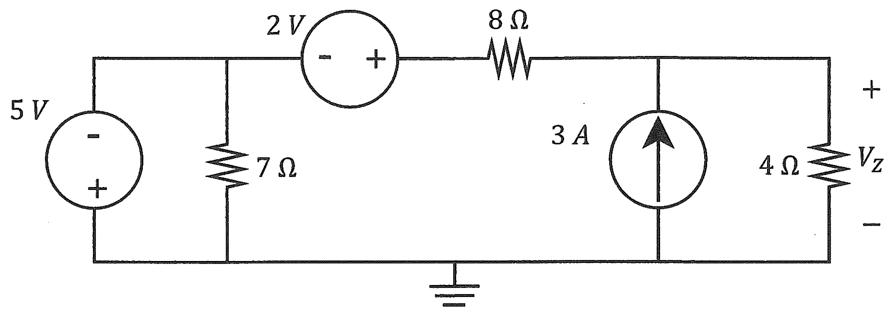
$12k\, i_2 = 5 + 3k(3m) + 5(-4m)$

$12k\, i_2 = 5 + 9 + 20$

$12k\, i_2 = -6$

$i_2 = \underline{-0.5mA}$ ✓

Vraag 3.3 [2] Gebruik node analise en bepaal V_Z.	**Question 3.3 [2]** Use nodal analysis to determine V_Z.



1

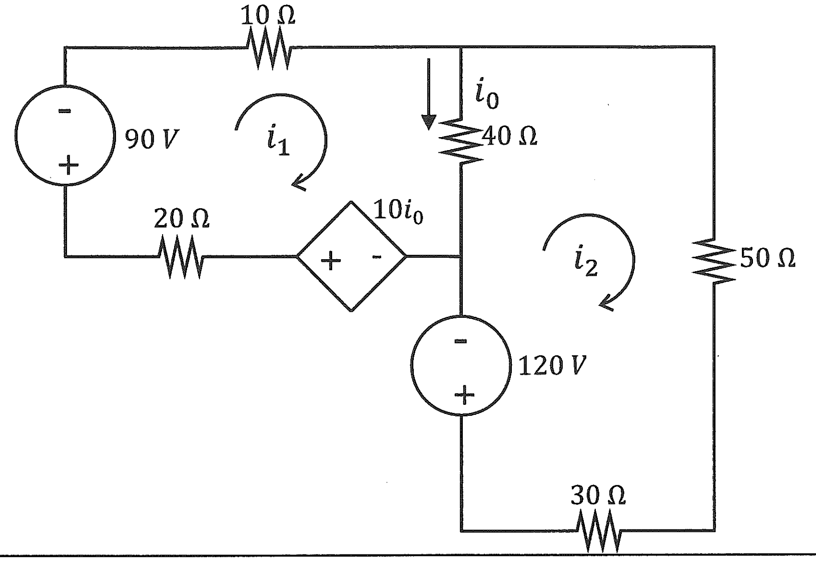
$$\frac{V_z}{4} + \frac{V_z - 2 - (-5)}{8} = 3.$$

$$2V_z + V_z + 3 = 24.$$

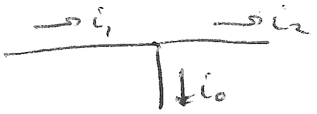
$$3V_z = 21$$

$$V_z = \underline{7V}$$ ✓✓

Vraag 3.4 [2]	Question 3.4 [2]
Indien gegee dat $i_2 = -1.8$ A, bepaal i_0 deur gebruik te maak van lusanalise.	If given that $i_2 = -1.8$ A, determine i_0 using mesh analysis.



①



$\therefore \; i_0 = i_1 - i_2.$

②

$$+90 + i_1(10+40+20) - 40i_2 - 10i_0 = 0.$$

$$90 + 70i_1 - 40i_2 - 10i_1 + 10i_2 = 0$$

$$90 + 60i_1 - 30(-1.8) = 0$$

$$60i_1 = -90 - 54$$

$$60i_1 = -144$$

$$i_1 = -2.4.$$

$$i_0 = -2.4 + 1.8$$

$$= -0.6 \text{ A}$$

Vraag 3.5 [4]	**Question 3.5 [4]**
a) Bepaal α indien $I_1 = 0.75$ A. $\begin{bmatrix} 4 & \alpha \\ -8 & 2 \end{bmatrix}\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 6 \\ -4 \end{bmatrix}$	a) Determine α if $I_1 = 0.75$ A. $\begin{bmatrix} 4 & \alpha \\ -8 & 2 \end{bmatrix}\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 6 \\ -4 \end{bmatrix}$
b) Bepaal β indien $v_1 = -3$ V en $v_3 = -1.8$ V. $\begin{bmatrix} 6 & 3 & -10 \\ -3 & 0 & \beta \\ 2 & -8 & 0 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} -3 \\ 0 \\ 2 \end{bmatrix}$	b) Determine β if $v_1 = -3$ V en $v_3 = -1.8$ V. $\begin{bmatrix} 6 & 3 & -10 \\ -3 & 0 & \beta \\ 2 & -8 & 0 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} -3 \\ 0 \\ 2 \end{bmatrix}$

a) $\Delta = 4\times 2 - (-8)\alpha = 8 + 8\alpha$

$\Delta_1 = \begin{vmatrix} 6 & \alpha \\ -4 & 2 \end{vmatrix} = 12 - (-4)\alpha = 12 + 4\alpha.$

$\forall \quad I_1 = 0.75 = \dfrac{12+4\alpha}{8+8\alpha} \Rightarrow 6 + 6\alpha = 12 + 4\alpha$

$2\alpha = 6$

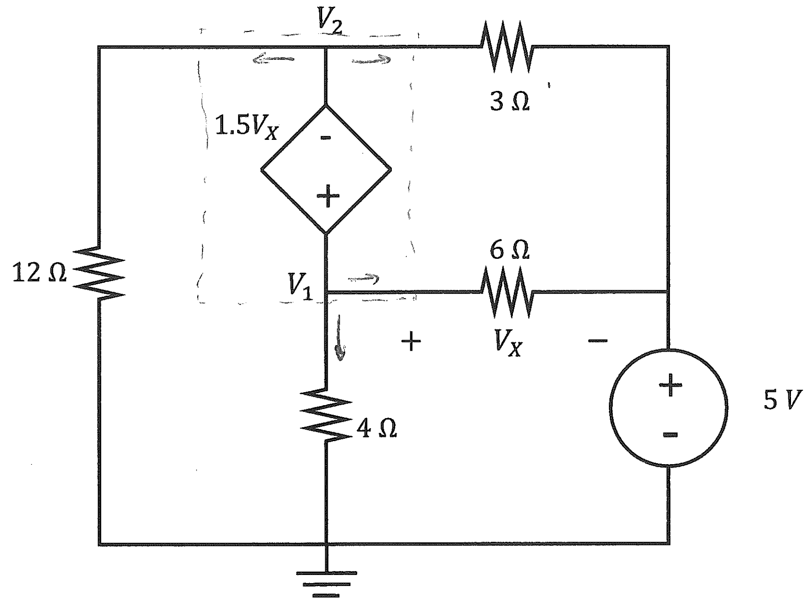
$\alpha = 3$ ✓✓

b) $-3v_1 + \beta v_3 = 0$

$-3(-3) + \beta(-1.8) = 0$

$\beta = {-9}/{-1.8} = 5$ ✓✓

Vraag 3.6 [4]	**Question 3.6 [4]**
Gebruik node-analise om twee vergelykings van die volgende vorm op te stel: $$aV_1 + bV_2 = 30$$ $$cV_1 + dV_2 = -7.5$$	Use nodal analysis to set up two equations of the following form: $$aV_1 + bV_2 = 30$$ $$cV_1 + dV_2 = -7.5$$



① $$\frac{V_1}{4} + \frac{V_1 - 5}{6} + \frac{V_2 - 5}{3} + \frac{V_2}{12} = 0$$

$$3V_1 + 2V_1 - 10 + 4V_2 - 20 + V_2 = 0.$$

$$\underline{\underline{5V_1}} \checkmark + \underline{\underline{5V_2}} \checkmark = 30$$

② $$V_1 - V_2 = 1.5V_x \quad \text{but } V_x = V_1 - 5.$$

$$\therefore V_1 - V_2 = 1.5V_1 - 7.5$$

$$\underline{\underline{-0.5V_1}} \checkmark \; \underline{\underline{-V_2}} \checkmark = -7.5$$